

Sovereignty Through Interdependence: Epistemic Agency, Intelligent Action, and Federated AI

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Abstract

Calls for Sovereign AI often focus on domestic ownership of data, compute, model weights, and deployment infrastructure. Recent scholarship and policy literature already show why full-stack autonomy is an incomplete objective, reframing sovereignty in terms of managed, strategic interdependence and national agency. Building on that premise, this position paper asks what such interdependence must preserve. We argue for two ends: (a) *epistemic sovereignty*, the legitimately governed capacity to shape how AI-mediated knowledge and understanding of a people are represented, validated, contested, and revised, both within AI knowledge stores and among people; and (b) *sovereignty of intelligent action*, the AI-mediated capacity of a population, and of the participating AI systems, for consequential goal-oriented action, together with the legitimate authority over how the goals, responsibility and accountability, and intervention and recourse associated with such action are allocated and delegated. We further describe *sovereignization* as the continuing process of creating, adapting, and continually augmenting participating AI systems in the service of stated sovereign ends. Sovereignty goals across individual, collective, and institutional units can conflict, requiring rights-preserving, contestable governance rather than *prima facie* priority for either global or local control. Federation is one important mechanism whereby a family of locally sovereign AI technologies can be created with *shared epistemic and action substrates*, leveraging shared information in open commons whenever positive cross-node transfer exists from non-poolable knowledge. We conclude with guidance on when shared, federated, local, retrieval-based, or excluded knowledge is appropriate: the resulting objective is not isolation but interdependence in which participating communities retain credible technical options, legitimate authority, and the capacity to keep learning.

CCS Concepts

• Human-centered computing → Collaborative and social computing; • Social and professional topics → Government technology policy; • Computing methodologies → Distributed artificial intelligence.

Keywords

AI Sovereignty, Digital Sovereignty, Epistemic Agency, Federated Learning, Pluralistic AI

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1 Sovereignty Beyond the AI Stack

Digital sovereignty is a broad phrase referring to state authority and strategic autonomy over digital infrastructure and platforms, data and computational systems and their impact on self-determination of activities for the national good [12, 19, 34, 37, 49]. In AI, the concentration of advanced accelerators, cloud infrastructure for large-scale frontier model training using large amounts of data (for serving inference and in-context and continuous training on the traces of deployed models), and scarce expertise — the last accruing to those already at the frontier — sharpens these concerns, producing structural dependencies in technology, data, and economics.

Recent work evaluates sovereignty across infrastructure, data, models, and governance, while warning that open weights alone do not remove structural dependence [13, 35, 41]. Public-sector work similarly treats redundancy, provider choice, retrieval, and auditability as components of institutional resilience rather than equating sovereignty with one domestically hosted model [27]. More broadly, a substantial recent literature rejects full-stack autarky outright. Singh and Sengupta [43] formulate AI sovereignty as a non-isolable continuum and argue for managed interdependence, while Tanner et al. [45] reach a similar conclusion after mapping transnational dependencies across the AI stack, and the World Economic Forum and Bain frame the objective as strategic interdependence [52]. Stanford HAI likewise argues that the operative question is which dependencies to re-balance for which goals, rather than whether the entire stack is domestically controlled [33]. The Tech Policy Design Institute goes further, recasting sovereignty as measurable AI agency — access, control, choice, and leverage —

across national capabilities [23], a re-framing also presented as a shift from sovereignty to agency in an interdependent system [36]. It is this move away from full-stack autarky, toward agency within interdependence, on which we build directly.

Our narrower question is: *what forms of agency must interdependence preserve, and through what continuing capabilities?* We draw on four bodies of work: digital sovereignty and technological dependence; epistemic agency and pluralistic alignment; human control and institutional authority over AI-mediated action; and open, distributed, and federated model development. We distinguish sovereign *ends* from their supporting *resources and mechanisms*. Our emergent position has three parts: sovereign ends (epistemic agency and authority over intelligent action), a continuing process (sovereignization), and an architecture combining common, shared, and local knowledge without a common endpoint. **AI sovereignty is the durable and legitimately governed capacity of a people, through contestable¹ institutions, to shape how AI-mediated knowledge and action are created, contested, authorized, and developed for its own legitimate good.**

We use sovereignty here in a functional *sociotechnical* sense: juridical sovereignty remains a property of sociopolitical orders, whereas communities and institutions exercise nested or derivative forms of agency within those orders. Infrastructure remains an important enabling condition, but it is not sufficient on its own: nominal authority without epistemic agency, scientific competence, operational capacity, or viable alternatives can leave choice largely formal. Conversely, dependence is not uniform, and external components can coexist with meaningful autonomy when effective structures and processes are in place to evaluate, modify, constrain, substitute, and govern their use. This qualification matters because infrastructural power can bound apparently free choices and become coercive under disruption [3, 18]. Effective agency [48, 54] therefore requires not only formal decision rights but also the credible capability and feasible alternatives that make those rights real.

2 Units of Agency, Knowledge, and Action

Agency is distributed across several levels. *Individuals* express and exercise choice within their rights and need access to recourse in case of conflicts. *Communities and institutions* exercise collective agency through language, culture, public rules, and governance. *Shared scientific communities* support open inquiry, interoperable evidence, common methods, and collaborative evaluation, resulting in a scientific commons — a shared resource regime. These levels are not legally co-equal sovereigns and can conflict: sovereignty typically concerns legitimate collective authority, while individual rights and shared epistemic standards constrain how those rights can be exercised. These conflicts run in several directions: collective alignment can suppress minority expression; individual action can limit collective learning; local norms can conflict with non-discrimination or due process; and universalized benchmarks can erase relevant linguistic or cultural context. Sovereignty and exercise of rightful agency require an ethical and institutional

framework preserving representation and participation, rights and non-discrimination, consent, contestability and recourse, and scientific integrity as applicable [8, 26, 33, 37, 44].

2.1 Epistemic Sovereignty

AI systems increasingly mediate search, summary, translation, recommendation, explanation, and knowledge production. Language technology can privilege dominant categories and contribute to epistemic injustice for underrepresented communities [5, 20, 24]. Empirical studies find measurable cultural skew in large language models, and culturally targeted prompting only partly corrects it [46, 50]. Recent work increasingly treats cultural and moral judgment as plural rather than as a single global alignment target [17, 38, 39, 53]. Branda and Cicciozzi [6] and Kakraba et al. [25] advocate *cognitive sovereignty* for retaining critical epistemic authority under AI-assisted cognition in science and health. Our emphasis is on the collective and institutional conditions of knowledge formation.

We use *epistemic sovereignty* for the collective capacity, under legitimate and contestable governance, to participate in determining how knowledge relevant to a community is sourced, validated, contextualized, represented, challenged, and revised. This extends work on epistemic agency in democratic life [10] and recent decolonial accounts of epistemic sovereignty [16]. Epistemic sovereignty is not epistemic isolation or relativism: shared evidence and scientific methods can coexist with agency over provenance, language, contextual interpretation, and contestation.

2.2 Sovereignty of Intelligent Action

As AI systems retrieve information, use tools, maintain state, recommend decisions, and execute actions, the issue extends beyond an epistemic concern into one of action. Meaningful-human-control research emphasizes traceability, responsiveness to human reasons, responsibility, and intervention [7, 14, 42]. Recent work on human oversight distinguishes AI operative agency from human evaluative agency [55], while work on agentic AI shows that controllability must address both causal intervention and normative authority at the system level [32].

We use *sovereignty of intelligent action* for both the property of the AI-mediated workflows to enhance the agency of the people they are expected to serve, and the higher-level institutional property that constitutes the legitimate self-governed authority to determine how objectives, permissions, delegation, intervention, responsibility, and recourse are allocated across the complete human-AI system. Meaningful human control asks whether humans can exercise meaningful control over an autonomous system; sovereignty of intelligent action additionally asks whether the authority to design and revise the institutional allocation of that control resides in a people.

3 AI Sovereignization as a Deliberate Process

A model need not be developed domestically from first principles to participate in a sovereign AI system. Open collaborative efforts such as BLOOM and OLMo show that globally produced foundation models can be public scientific artifacts, with OLMo exposing training data, code, evaluations, and intermediate artifacts for scrutiny [21, 51]. Domain-adaptive pretraining can reshape broad models

¹We use *contestability* in the sense developed in the contestable-AI literature (notably, Almada [2] and Alfrink et al. [1]): the capacity of those affected by an AI-mediated decision, knowledge claim, or action, to challenge it and obtain explanation, human review, and — where warranted — revision. It is a property of the sociotechnical system and its governance, distinct from the term's specialized senses in other fields.

Table 1: Federation of shared-private knowledge among sovereign nodes for enabling AI sovereignty for all.

Condition (Information/Governance)	Primary route
Poolable and broadly useful	Open/Common or centralized training
Non-poolable with empirically positive cross-node transfer	Federation or selective model/artifact exchange
Predominantly local and non-transferable	Local adaptation, private modules, or local memory
Volatile or institutionally authoritative	Retrieval, database, verifier, or tool
Prohibited, unsafe, or of little or no value	Exclude from the learning path

using local corpora [22], while retrieval can keep mutable or authoritative knowledge outside model parameters [28]. We use *AI Sovereignization* for the continuing acquisition and exercise of the ability to independently inspect and evaluate, replace, constrain and adapt possibly global AI components under legitimate local governance. The term has previously been used for territorial or state-control projects concerning Internet infrastructure [30]; here it denotes a capability-centered process compatible with cross-border collaboration.

Openness should be disaggregated. *Open weights* can improve local deployment and modifiability; *open model development* additionally exposes code, data and training information; *portable system architecture* limits dependence across model, retrieval, tool, identity, evaluation, observability, and serving interfaces. None alone guarantees sovereignty [41]. Consider a public-interest institution deploying a multilingual public-service assistant. There, sovereignization is the continuing ability to evaluate the assistant on local languages and customs, identify and eliminate failure modes, update local knowledge and usage policy, constrain permissions, and migrate to other base models when needed. A nominal right to switch is insufficient if the alternative is technically or economically infeasible [3].

4 Federation for Sovereignty and Interdependence

The concentration of frontier capability in a few heavily funded labs and countries (and the widening expertise gap it produces) is a sovereignty problem as much as a scientific one. Open, federated development initiatives offer a response to the above challenges by creating shared ecosystems. Federation amortizes training cost across distributed compute clusters while pooling locally held knowledge that no single node could assemble on its own. Classical federated learning learns from decentralized data without moving raw records [31], and personalized methods permit local models to differ when distributions or objectives differ [11, 29], similar to privacy-enhancing/split-learning collaboration in public agencies [9]. Low-communication approaches show that language-model training spans weakly connected compute islands, including cross-silo pretraining at billion-parameter scales [15, 40].

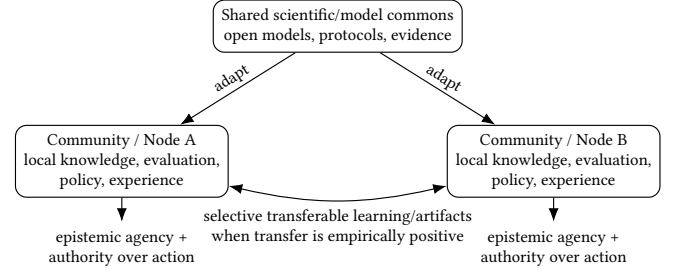
**Figure 1: Agency-preserving interdependence permits shared foundations and selective transfer without identical local endpoints.**

Table 1 presents the governance conditions for sovereign nodes sharing data and information with each other, allowing for the creation of AI technologies that preserve AI sovereignty for all. Federation is technically warranted when controlled evaluation establishes positive cross-node transfer from non-poolable information and that value exceeds communication, optimization, privacy, governance, and coordination costs (Figure 1). Conversely, information that is unique to one node may legitimately remain in local adapters, retrieval systems, tools, or institutional procedures.

Project Tapestry is a contemporary public example of this design direction: a consortium approach in which participants retain local data and compute while collaboratively improving open foundations and developing derivatives [47]. It is one possible realization, not evidence that federation is always required; local adaptation, retrieval, secure execution, distillation, or selective artifact exchange may be preferable in other regimes.

Federating the Example. Now consider a federation of such public-interest institutions, in several countries or regions, building multilingual public-service assistants. Rather than each node attempting to reproduce a frontier model independently, the participants begin from a jointly accessible open foundation and contribute complementary resources: compute clusters, language and domain expertise, locally available data, evaluation capacity, and engineering effort. Shared training and experimentation amortize costs and allow researchers across the federation to participate directly in frontier model development, while the resulting models, methods, and evaluation assets remain open to all participating nodes. Capabilities that transfer across nodes — cross-lingual representations for shared morphosyntactic architecture across the Indic language family [4], for instance — can improve the common substrate through federated learning without raw data leaving its originating institution, whereas intrinsically local information, such as jurisdiction-specific service rules or sensitive interaction histories, stays in local models, retrieval systems, and institutional procedures (Table 1).

5 Conclusion

Recent work has already moved AI sovereignty away from literal self-sufficiency toward managed interdependence, strategic choice, and agency. This paper develops the next layer: it distinguishes the sovereign ends that interdependence must preserve—*epistemic*

sovereignty and *sovereignty of intelligent action*—treats sovereignty as a continuing capability rather than a property of model provenance, and identifies when shared, federated, local, retrieval-based, or excluded knowledge is appropriate. As these ends are necessarily plural and may be contentious across individual, collective, and scientific institutions, the objective is not isolation but interdependence among communities that retain credible technical options, legitimate authority, and the ability to keep learning.

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